From Data to Decisions: Client Churn Analysis in a Digital Finance Platform

Business understanding · Data cleaning · Exploratory data analysis · Descriptive analytics · Machine learning proposal & implementation

Student Numbers: 3602767, 3602738, 3602737

5,000
Clients

20.7%
Churn rate

Orange
Implementation

Business Understanding

The client is a digital financial services provider facing client churn, where previously active customers stop using the platform and no longer generate revenue. Predicting churn is important so the company can identify at-risk clients early and target them with retention actions such as offers, better support, or product changes.

Churn is represented by **Left_Service** in the dataset. Key variables likely to influence churn include: **Membership_Length** (relationship duration), **Account_Value** and **Product_Count** (value and engagement), **CreditCard_Flag** and **Premium_Tier** (deeper commitment and higher revenue potential), **Activity_Status** (early sign of disengagement), **Risk_Score** and **Annual_Earnings** (profitability and sensitivity to pricing/credit), and demographics like **Age**, **Gender**, and **Region** (which may reveal high-risk segments).[2]

Data Preparation

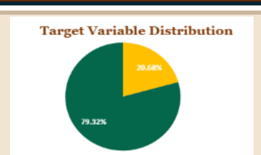
- 5 missing and 21 duplicate values were found and handled.
- Standardised categorical labels and created grouped fields (Age_Group, Balance_Band, Income_Band, Membership_Band, Income_Band) to make churn patterns easier to interpret.
- Initially there were 5,031 rows; after removing problematic or duplicate records, the final dataset contains 5,000 unique clients with consistent, analysis-ready variables. [1]

Issue Checked	How detected	Result/ Action
Missing values	Highlight missing values, COUNTBLANK per column	Annual_Earnings-1, replaced with average. Product_Count-1, replaced with mode. CreditCard_Flag- 2, replaced with mode. Activity_Status- 1, replaced with mode.
Duplicated values	Remove duplicates on ClientID as a unique identifier	21 duplicate values were found and removed.
Inconsistent categories	Filter unique values per categorical var	Standardised label for Gender-Male/Female, before few where M and F
Outliers	Summary statistics, using IQR values and box-plots	Annual_income-3 negative values were outliers (value<-98454.7713). Age- 376 outliers, removed outliers (6,9,9) and kept outliers in range 57-84. Membership_Length- 3 outliers (values>13) were removed Account_value- 3 outliers (values>319428.4) were removed.

Original field	Example value	Prepared field	Example value	Reason for transformation
Risk_Score	601	Risk_Band	600-699	Groups similar score together
Age	52	Age_Group	50-59	Easier to compare churn by age segment
Account_Value	64430.06	Balance_Band	25k-75k	Groups similar balances together
Membership_Length	2	Membership_Band	1-3	Simplifies churn comparison by years
Annual_Income	96517.97	Income_Band	50k-100k	Groups similar income values together

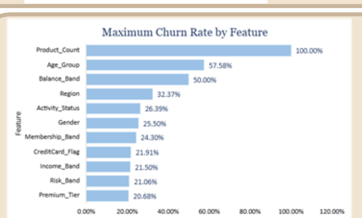
Data cleaning and transformation summary

Exploratory Data Analysis



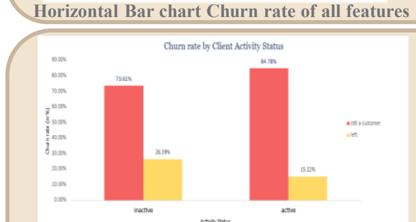
This chart shows that most of the customers are still present and only a few have left the service, this helps the business to understand what portion of customers may have an issue or we have to deal with.

Pie chart of Left_Service (left vs still a customer).



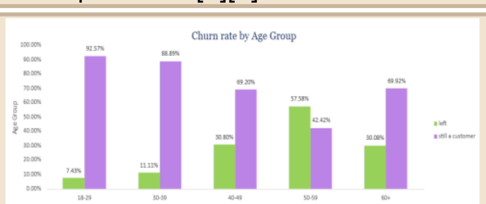
Horizontal Bar chart Churn rate of all features

- We use this chart to screen which features show the largest churn variation.
- Although some variables (e.g. Balance_Band) have a high maximum churn in small subgroups, we selected **Age_Group, Product_Count, and Activity_Status** as primary EDA focus because they combine strong churn differences with large group sizes and clear business interpretation.[3][1]



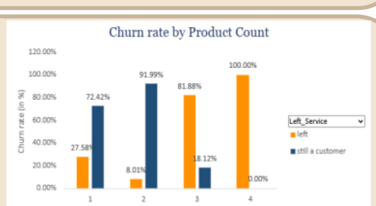
Client Activity Vs Churn (Grouped bar chart)

This chart shows that customers with active status have lower churn rate than customers having inactive status, which tells that keeping customer engaged is crucial to minimize churn rates.



Churn rate by Age group (Grouped bar chart)

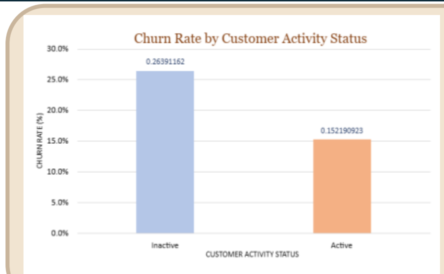
The chart shows that young customers are more engaged and have lower churn rates than older customers, which suggests that special attention should be given to middle-aged clients.



Churn rate by Products (Grouped bar chart)

The chart shows that customers few products have a lower churn rate while customers with many products are more likely to leave, which could be due to issues in multi-product package services.

Descriptive Analytics



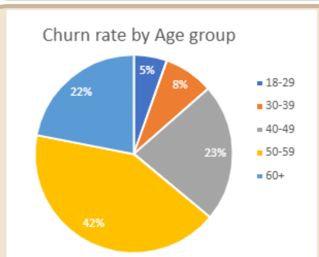
Customer Activity Status Bar Chart

1. Are inactive customers more likely to leave the Service?

Yes. Inactive customers churn at a much higher rate than active customers, and above the company average.

What this means: Inactivity is a warning sign. Reaching out early can prevent churn.

What the client can do: Track prolonged inactivity and send nudges or small incentives to re-engage users.



Churn Rate by Age Group Pie Chart

2. Which age group churns the most?

The oldest age group churns the highest, followed by the middle age group. The youngest group churns the least, well below the company average.

What this means: Older customers are leaving at much higher rates — something about the product is failing this demographic.

What the client can do: Investigate why older customers leave (features, technical issues, alternatives) and tailor retention efforts to them.[1]

Machine Learning Proposal and Implementation

This is a **binary classification problem** because the target variable, Left_Service, has two outcomes: left or still a customer. Based on the cleaned dataset, the input features are customer demographic, behavioural, and financial variables such as Age, Membership_Length, Product_Count, Activity_Status, Risk_Score, Annual_Earnings, Account_Value, Gender, Region, and CreditCard_Flag.

Random Forest model is suitable because it can handle both numerical and categorical data, capture complex non-linear relationships, and is widely used in customer churn prediction studies. It is also robust and provides strong predictive performance on structured datasets like this one.[1][4]

Paper	Algorithm	Evaluation	Reason
1. Anusha & Keerthi (2025), Bank Customer Minimization Churn Rate Using Machine Learning	AdaBoost	88–92%, high recall	Performs best due to boosting and handling class imbalance
	Random Forest	80–88%	Strong model but slightly lower recall than AdaBoost
	Decision Tree	70–80%	Simple model but prone to overfitting
2. Ashraf (2024), Bank Customer Churn Prediction Using Machine Learning Framework	Random Forest	85–90%	Performs consistently well on structured banking data
	XGBoost	88–93%	Captures complex patterns and high predictive power
	AdaBoost	85–90%	Boosting enhances model performance by focusing on difficult cases
	Bagging Meta	90–95%	Highest stability and accuracy
3. Chen (2024), Credit card customers churn prediction by nine classifiers	Random Forest	85–90%	Top-performing classifiers
	Decision Tree	70–80%	Lower performance
	Support Vector Machine	80–88%	Performs well but less flexible
4. Orina, Rimiru & Mwangi (2023), A Comparative Study of	Random Forest	85–90%	Strong performance
	Support Vector Machine	80–85%	Good but less stable

Summary of research papers on bank client churn predictions.

Final Model and Results

The dataset was analysed in **Orange** using selected input features and a Random Forest classifier, evaluated with 10-fold cross-validation. The results show that machine learning can effectively predict customer churn and support proactive retention efforts.

Actual \ Predicted	left	still a customer	Σ
left	449	585	1034
still a customer	180	3786	3966
Σ	629	4371	5000

Confusion Matrix for Random Forest from Orange

Model	AUC	CA	F1	Prec	Recall	MCC
Random Forest	0.818	0.847	0.832	0.835	0.847	0.475

Test Metrics for Random Forest from Orange

Model	Test and Score
Logistic Regression	Recall - 80.7% Precision - 77.6%
Random Forest	Recall - 84.7% Precision - 83.5%
AdaBoost	Recall - 79.3% Precision - 79.4%

Evaluation metrics for the top 3 suggested models.

Appendix & Acknowledgement

Additional details on data preparation, exploratory data analysis, and machine learning research supporting the results shown in this poster are available in the accompanying Excel workbook prepared for this project.

References

[1] J. Adeojo, "Using Machine Learning to Predict Customer Churn | Towards Data Science," *Towards Data Science*, Feb. 14, 2021. https://towardsdatascience.com/using-machine-learning-to-predict-customer-churn-cd499cb230db/

[2] "Causes of customer churn | Stripe," *Stripe.com*, 2024. https://stripe.com/ae/resources/more/what-causes-churn-and-how-businesses-can-minimize-it

[3] S. Charandabi, "Prediction of Customer Churn in Banking Industry." Available: https://arxiv.org/pdf/2301.13099

[4] SCISPACE, "SciSpace," *scispace.com*, 2025. https://scispace.com/